

Course Code	Course Title	L	T	P	C
BCSE408L	Cloud Computing	3	0	0	3
Pre-requisite	NIL	Syllabus version			
		1.0			
Course Objectives					
1. To understand the fundamental of cloud computing and the virtual machine 2. To gain knowledge of the various cloud service and deployment models 3. To understand cloud management and cloud security concept					
Course Outcomes					
Upon completion of the course, the student will be able to 1. Design and develop cloud application and deploy it. 2. Evaluate the various cloud services and deployment models in the infrastructure 3. Apply the various cloud security concepts for application development 4. Design and manage cloud services with cloud simulation and various cloud platforms. 5. Design and develop AI and IoT applications in the cloud environment					
Module:1	Introduction	5 hours			
Cloud Computing definition - Evolution of Cloud Computing - Benefits and challenges of cloud computing – Cloud services – Cloud deployment – Cloud architecture –NIST architecture - Business models.					
Module:2	Virtualization	6 hours			
Introduction to Virtual Machine (VM) - basics of Virtualization - Types of Virtualizations - Desktop Virtualization – Application Virtualization - Server Virtualization - Storage Virtualization- OS level Virtualization –Virtualization for cloud computing – Software-defined data Center (SDDC).					
Module:3	Public Cloud	7 hours			
Public cloud benefits – Challenges – public cloud services – AWS – compute – storage –network services –Google cloud service (GCP) – compute – storage – network – Cloud AI services – Multitenant - case study.					
Module:4	Private Cloud	7 hours			
Private cloud benefits – challenges – private cloud services – VM migration – cloud provisioning – managing private cloud - OpenStack architecture – components – OpenStack installation –Google private cloud services - case study.					
Module:5	Cloud Management & Security	6 hours			
Data center –cloud management – resource management - automation –benefits of automation - Infrastructure security – network security – host level security.					
Module:6	Security Principles	6 hours			
Cloud security overview – CIA triads - Threats – risk management - computer security incident response team (CSIRT)–cloud security design principles - cloud security standards: privacy, confidentiality, and integrity –cloud security policy – service level agreement (SLA)					
Module:7	Cloud Application development	6hours			
Tools for cloud development – simulators – cloudsim - develop an application and deploy in public cloud services – deploy AI application in the cloud – IoT cloud services – cloud security services.					
Module:8	Recent Trends	2 hours			